

claude-windows / 7d6bbb83-42c0-459b-9e23-e14e96d4cb2d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7d6bbb83-42c0-459b-9e23-e14e96d4cb2d\subagents\workflows\wf_1dc2acd6-052\agent-ae15fb17e83823554.jsonl`  
- SHA-256: `993dcdcb9fdd2b534f1b6b4c5cd623176f3d4167c17f79c8cccfefebe1026436`  
- Source modified: `2026-06-13T20:23:47+00:00`  
- Imported at: `2026-07-08T16:00:13+00:00`  
- project: `wf_1dc2acd6-052`  
- session_id: `7d6bbb83-42c0-459b-9e23-e14e96d4cb2d`
```

Transcript

1. user (2026-06-13T20:19:21.972Z)

You are an ADVERSARIAL fact-checker. Theme: "Cascades / two-stage proposal-verify; decouple detection from decision; offline re-scoring & feature caching".

Here are a teammate's survey findings (JSON):

```
{  
  "theme": "Cascades / two-stage proposal-verify; decouple detection from decision; offline re-scoring & feature caching",  
  "our_pattern": "\"Persist once, re-score forever\": expensive per-frame perception (a frozen shuttle-tracking CNN like WASB/TrackNet, a person detector, an audio thwack detector, a proposed per-frame shuttle-state classifier) runs ONCE over the video and is persisted as time-aligned signals. The cheap DECISION layer ? windowing into candidate rally intervals + a tiny logistic-regression scorer over per-window feature columns ? is a pure, deterministic, offline re-scoring step. Most experiments re-score stored features and never touch the served perception pipeline. Detection ? decision is the architectural axis; cues are (value, confidence) feature columns the scorer learns to weight (no hand-coded if/else in the decision path); reconcilers are a report-first post-hoc pass that flags where the decision contradicts persisted evidence; everything is graded offline with leave-one-video-out vs scarce golden intervals using temporal IoU, flag-gated default-OFF, promoted only on measured lift.",  
  "key_papers": [  
    {  
      "title": "Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs (Segment-CNN / S-CNN)",  
      "authors": "Zheng Shou, Dongang Wang, Shih-Fu Chang",  
      "year": "2016",  
      "url_or_id": "arXiv:1601.02129 (CVPR 2016)",  
      "venue": "CVPR 2016",  
      "what_it_does": "The canonical two-stage temporal-action-localization framework: a PROPOSAL network filters background segments, a CLASSIFICATION network is trained one-vs-all, and a LOCALIZATION network refines/scores instances. Reports mAP at temporal IoU. Establishes proposal->classify->localize decoupling for untrimmed video, the direct analog of our window-then-score split.",  
      "overlap": "strong",  
      "overlap_explanation": "Same core structure as ours: a cheap proposal/candidate-window stage decoupled from a downstream scoring/classification stage, evaluated at temporal IoU on untrimmed video. Differences: S-CNN uses 3D-ConvNets end-to-end and is NOT 'persist perception once, re-score many'; ours freezes perception and re-scores a tiny LR. But the proposal/decision decoupling and the tIoU metric are exactly the established framing we are using."},  
    {  
      "title": "Cascaded Boundary Regression for Temporal Action Detection (CBR)",  
      "authors": "Jiyang Gao, Zhenheng Yang, Ram Nevatia",  
      "year": "2017",  
      "url_or_id": "arXiv:1705.01180 (BMVC 2017)",  
      "venue": "BMVC 2017",  
      "what_it_does": "Two-stage cascade: stage 1 detects class-agnostic proposals, stage 2 detects specific actions; temporal boundaries are refined in a cascaded way by feeding refined windows back for further regression. Reports tIoU/mAP on THUMOS-14.",  
      "overlap": "strong",  
      "overlap_explanation": "Directly the 'cascade / two-stage proposal-verify + boundary refinement' pattern. Its cascaded boundary regression is the learned analog of our reconcilers that tighten segment boundaries to serve-contact/inactivity. Difference: CBR's refinement is a learned regressor inside the model; our boundary tightening is a report-first rule pass A/B-gated before auto-apply."},  
    {  
      "title": "Decoupled Classification Refinement: Hard False Positive Suppression for Object Detection (DCR)",  
      "authors": "Bowen Cheng, Yunchao Wei, Rogerio Feris, Jinjun Xiong, Wen-mei Hwu, Thomas Huang, Honghui Shi",  
      "year": "2018",  
      "url_or_id": "arXiv:1810.04002",  
      "venue": "ECCV 2018 / arXiv",  
      "what_it_does": "Adds a STANDALONE classifier that re-scores a base detector's boxes to kill HARD FALSE POSITIVES, deliberately NOT sharing features with the detector. At inference all
```

detections are rescored by the decoupled head.", "overlap": "strong", "overlap_explanation": "This is the cleanest published statement of 'decouple a cheap detector from a separate verify/rescore stage to suppress hard false positives,' which is exactly what our LR scorer + reconcilers do to candidate windows (drop zero-net-crossing segments, etc.). Spatial object detection vs our temporal rallies, but the decouple-then-rescore principle and the 'hard false positive' framing are identical."}, {"title": "Rapid Object Detection using a Boosted Cascade of Simple Features (Viola-Jones rejection cascade)", "authors": "Paul Viola, Michael Jones", "year": "2001", "url_or_id": "CVPR 2001 (doi:10.1109/CVPR.2001.990517)", "venue": "CVPR 2001", "what_it_does": "The seminal attentional/rejection cascade: cheap early stages reject most non-object windows so expensive computation is spent only on promising candidates; later stages are progressively more complex.", "overlap": "partial", "overlap_explanation": "Establishes the foundational vocabulary ('cascade', 'rejection/attentional cascade', 'cheap-reject-then-expensive-verify'). We use the inverse compute ordering (expensive perception first, persisted; cheap decision second), so it is the conceptual ancestor rather than a structural twin ? overlap is on terminology and the staged compute-budget idea, not the exact pipeline."}, {"title": "Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from a Monocular Camera", "authors": "Yi-Hua Hsu, Chih-Chang Yu, Hsu-Yung Cheng", "year": "2024", "url_or_id": "Sensors 24(13):4372, doi:10.3390/s24134372 (PMC11244353)", "venue": "Sensors (MDPI) 2024", "what_it_does": "Fuses TrackNet shuttle trajectory + YOLOv7 swing/hit detection; a rule-based Shot Refinement Algorithm (SRA) reconciles the two streams via 5 explicit decision cases (accept hit, reject trajectory in non-swing segment, dedupe multiple detections, recover missing trajectory from swing, confirm true negative). Lifts shot accuracy from 58.8% (TrackNet alone) to 89.7%.", "overlap": "strong", "overlap_explanation": "Same domain AND same reconcile-fused-detector-outputs idea as our reconcilers: cross-checking the shuttle stream against the person/swing stream to correct false positives/negatives at segment boundaries. Difference: SRA hard-codes if/else cases (exactly the 'special-casing' we deliberately avoid in the decision path), and it is shot-level not rally-level. Strong evidence our reconciler instinct is sound, and a cautionary example of the hand-coded-branch path we are trying to avoid in the scorer."}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "authors": "Zhaoyu Liu, Xi Weng, Lianyu Hu, Zhe Hou, Kan Jiang, Jin Song Dong, Yang Liu", "year": "2026", "url_or_id": "arXiv:2603.13397", "venue": "arXiv 2026", "what_it_does": "To segment full match broadcasts into rally clips it runs a lightweight purely-audio racquet-impact detector (augmented with crowd/announcer noise), CLUSTERS hit moments into candidate rally intervals, then FILTERS by view/scoreboard constraints. Decouples cheap detection from downstream validation.", "overlap": "strong", "overlap_explanation": "Almost exactly our pipeline shape in a sibling sport: cheap per-frame/audio detector -> cluster into candidate intervals -> rule-filter/decision. Confirms that 'audio thwack detector -> candidate windows -> decision filter' is established practice for racket-sport rally segmentation. Difference: their decision is a fixed filter; ours is a learned scorer + LOVO-graded harness."}, {"title": "A New Perspective for Shuttlecock Hitting Event Detection", "authors": "Yu-Hsi Chen", "year": "2023", "url_or_id": "arXiv:2306.10293", "venue": "arXiv 2023 (CoachAI/badminton challenge)", "what_it_does": "Detects shuttlecock hitting events from trajectory; representative of the badminton trajectory->event pipeline that underpins rally/shot structuring from TrackNet-style per-frame outputs.", "overlap": "weak", "overlap_explanation": "Confirms the broader practice of deriving discrete events/decisions from persisted per-frame shuttle trajectory (our detection->decision split at the cue level), but it targets hit-event detection rather than rally-interval segmentation or a re-scorable decision layer, so overlap is thematic/weak."}], "established_terms": ["two-stage detection (proposal generation then classification)", "cascade / rejection (attentional) cascade", "temporal action proposal generation", "temporal action localization / detection (TAL/TAD)", "actionness score (candidate-likelihood per snippet/window)", "decoupled classification refinement / re-scoring head", "hard false positives", "cascaded boundary regression / boundary refinement", "over-segmentation error", "temporal Intersection-over-Union (tIoU) and mAP@tIoU", "frozen-backbone precomputed (cached) features", "leave-one-video-out / grouped cross-validation; video-level split", "off-policy / offline policy evaluation (replay)"], "techniques_to_adopt": [{"technique": "Adopt and report mAP@tIoU at multiple thresholds (e.g. 0.3/0.5/0.7) and average-mAP, not just a single mean temporal IoU.", "why": "This is THE metric the temporal-action-detection field (S-CNN, CBR) expects; reporting it makes your results comparable/legible to that literature and exposes whether gains are at loose or tight boundaries (where reconcilers act).", "effort": "low", "from_paper": "Temporal Action Localization via Multi-stage CNNs (S-CNN, Shou 2016); Cascaded Boundary Regression (Gao 2017)"}, {"technique": "Frame the candidate-window stage explicitly as 'temporal action proposal generation' and emit an 'actionness'/proposal-confidence per window that the scorer consumes.", "why": "Aligns your windowing with established proposal-generation literature, gives a principled recall-vs-#proposals knob (AR@AN), and lets

you measure proposal recall separately from decision precision ? diagnosing whether misses are a windowing or a scoring failure.", "effort": "medium", "from_paper": "S-CNN (Shou 2016); BSN/two-stage TAD proposal literature"}, {"technique": "Mine and explicitly train the scorer on the detector's HARD FALSE POSITIVES (windows that look rally-like but are not), as a decoupled re-scoring head.", "why": "DCR shows that in decoupled detection most remaining errors are hard false positives from the classification (decision) side, fixable by a separate rescorer fed hard negatives ? directly applicable to your LR scorer and to choosing which reconciler rules matter most.", "effort": "medium", "from_paper": "Decoupled Classification Refinement (Cheng/Shi 2018)"}, {"technique": "Make boundary tightening a learned/iterative regression (cascaded boundary regression) rather than only fixed rule precedence, or at least benchmark your rule-based reconcilers against a CBR-style learned refiner.", "why": "CBR's cascaded boundary regression is the established way to refine temporal boundaries and gave large mAP@0.5 gains; it is the learned counterpart to your truncate/split/tighten reconcilers and a natural A/B baseline to justify staying rule-based.", "effort": "high", "from_paper": "Cascaded Boundary Regression (Gao 2017)"}, {"technique": "Treat the over-merged-segment and over-fragmented-segment cases as the named 'over-segmentation' problem and consider a boundary-detection branch / ASRF-style refinement as a learned alternative to the rule-based split/merge reconcilers.", "why": "ASRF formalizes correcting over-/under-segmentation with a dedicated boundary branch that refines a frame-wise classifier ? exactly your split-on-internal-dead-time and merge reconcilers ? giving you a citable method and a learned baseline to compare against.", "effort": "medium", "from_paper": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF, Ishikawa 2020)"}, {"technique": "Keep the SRA-style cross-stream reconciliation but express each of its decision cases as FEATURE COLUMNS for the scorer rather than hard if/else, and cite SRA as the rule-based baseline you measure lift against.", "why": "The badminton SRA proves cross-checking shuttle vs swing streams works (58.8%->89.7%), but its hard-coded 5-case logic is precisely the special-casing you forbid in the decision path; turning those cases into columns honors your 'signal not verdict' rule while borrowing the proven cues.", "effort": "medium", "from_paper": "Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection (Hsu 2024)"}, {"technique": "Adopt offline/off-policy 'replay' evaluation framing for your re-scoring experiments and state estimator variance, since the golden set is tiny.", "why": "Your 'persist once, re-score forever, promote on measured lift' loop IS off-policy/offline policy evaluation; using its vocabulary (replay, estimator variance, confidence intervals) guards against over-trusting a lift measured on ~6 matches and connects you to a rigorous literature.", "effort": "low", "from_paper": "Off-policy / offline policy evaluation literature (Benchmarks for Deep Off-Policy Evaluation, arXiv:2103.16596; offline replay evaluation)"}, {"pitfalls": ["Proposal-recall ceiling: in two-stage pipelines the decision stage can never recover rallies the windowing missed. If your candidate-window generator drops a real rally, no LR re-scoring fixes it ? measure proposal/window recall (AR@AN) separately, or you will misattribute recall losses to the scorer (S-CNN/TAD literature).", "Decision-side hard false positives dominate once perception is decoupled (DCR): the biggest remaining errors are non-rally windows that look rally-like; a scorer not trained on mined hard negatives will plateau. Class imbalance (dead-time >> rally windows) makes a plain LR over-predict the majority unless reweighted/calibrated.", "Over-segmentation / over-merging is a named, persistent failure of cheap segment decisions (ASRF): long static-camera rallies get split at momentary occlusions and short dead-time gaps get merged; rule-based smoothing can over-correct and erase true short rallies. A/B every reconciler for net lift, not just count of 'issues fixed'.", "LOVO leakage with a static-tripod corpus: grouping by match is necessary but may be INSUFFICIENT ? same venue/court/camera/players across matches still leak scene/appearance cues into a frame-state classifier (video/scene-level split warnings). With ~6 matches each fold is high-variance; a single match can swing the lift sign.", "Tiny-corpus / off-policy variance: a lift measured on ~6 vendor-labeled matches has wide confidence intervals; 'promote only on measured lift' can promote noise. Use confidence intervals / per-match breakdown and beware multiple-comparisons as you flag-gate many cues.", "Audio-cue brittleness for the thwack detector (your own E1 finding, echoed by TennisExpert's need for heavy noise augmentation): impulsive crowd/announcer noise yields high recall but poor discrimination; clustering hits into intervals inherits this and can fragment or over-extend candidate windows.", "Calibration drift across decoupled stages: persisted detector confidences are not probabilities the LR can trust as-is; without per-cue calibration the learned weights conflate 'cue is confident' with 'cue is informative,' undermining the 'signal not verdict' goal.", "Persist-once staleness: 'persist once, re-score forever' silently couples every downstream experiment to a fixed detector version; if you upgrade WASB/TrackNet or the person detector you must invalidate and recompute the cache or experiments compare against stale evidence (feature-store versioning discipline)."], "verdict": "partial-overlap", "verdict_rationale": "The individual architectural moves are well-established and should be cited rather than presented as novel: two-stage proposal-then-classify (S-CNN), cascaded boundary refinement (CBR), decoupled re-scoring to kill

hard false positives (DCR), the cascade/rejection lineage (Viola-Jones), over-segmentation boundary refinement (ASRF), and even the exact racket-sport pipeline shape (TennisExpert: audio hits -> cluster -> candidate intervals -> filter; SRA: cross-stream rule reconciliation in badminton). So the 'cascade / decouple detection from decision' theme on its own is reinventing known ideas. What is NOT a verbatim match in the literature found is the SPECIFIC combination: (a) elevating detection-decision decoupling to a persisted 'compute once, re-score many' substrate where nearly all experiments are pure offline re-scoring of frozen per-frame evidence, (b) a strict 'signal not verdict / no special-casing ? every cue is a feature column, the model learns weights' discipline that deliberately rejects the hand-coded SRA-style branch logic that the closest domain papers actually use, and (c) report-first, A/B-lift-gated, idempotent RECONCILERS treated as a linter over decisions vs evidence rather than a learned in-model refiner. That packaging ? TAD-style decoupling + feature-store-style caching + off-policy lift-gated promotion + a report-first reconciler linter, on a scarce LOVO-graded racket-sport corpus ? is a sensible novel combination of established parts, not a research gap. Hence partial-overlap: borrow the field's metrics (mAP@tIoU, AR@AN), vocabulary (proposals/actionness, hard false positives, over-segmentation), and learned baselines (CBR/ASRF) as course corrections, while your contribution is the engineering discipline around them.", "search_queries_used": ["cascade object detector two-stage proposal then classify Viola Jones rejection cascade", "temporal action detection proposal generation then classification two-stage video", "decouple perception from policy video pipeline precompute features offline rescoring", "off-policy offline evaluation reranking cached features compute once decide many", "badminton rally detection segmentation broadcast video play break shuttlecock TrackNet", "tennis match video temporal segmentation rally point detection from broadcast audio visual cues", "cascade temporal action detection coarse to fine boundary refinement reduce false positives", "SCNN segment CNN temporal action localization untrimmed video proposal classification localization", "feature store precompute embeddings train and rescore video frozen backbone offline reproducible experiments", "rule-based post-processing temporal action segmentation refinement merge split over-segmentation correction", "leave one group out cross validation small dataset video data leakage temporal evaluation pitfall", "computation reuse video inference caching intermediate features avoid recompute query engine"]}]

Your job is to be skeptical and protect against two failure modes: (1) FABRICATED citations, (2) OVERCLAIMED novelty.

DO, using WebSearch/WebFetch (load via ToolSearch "select:WebSearch,WebFetch" if needed):

1. For each cited paper, search its EXACT title (+authors) to confirm it is a REAL paper. Mark verified_real
yes/no/uncertain and flag any that don't resolve in fabrication_flags. Be strict: a plausible-sounding title that you cannot find is a fabrication flag.
2. For every "novel-combination" or "genuine-gap" verdict, actively try to REFUTE it: run fresh searches for the
CLOSEST prior art that would downgrade the verdict toward "reinventing"/"partial-overlap". Default to assuming prior art exists until your searches come up empty. List any closer_prior_art you find (with real ids).
3. Give a corrected_verdict + rationale. Add extra_course_corrections if your searches surfaced adopt/avoid notes.

Return the structured object.

2. assistant (2026-06-13T20:19:24.351Z)

I'll fact-check this survey adversarially. Let me load the web tools and start verifying each cited paper.

3. assistant (2026-06-13T20:19:29.431Z)

I'll verify each of the 8 cited papers by searching for exact titles and authors, running parallel searches.

4. user (2026-06-13T20:19:38.825Z)

Web search results for query: "Temporal Action Localization in Untrimmed Videos via Multi-stage

Links: [{"title": "Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs | Scilit", "url": "https://www.scilit.com/publications/84bc9ef996b86e0d33c9405deb241ca0"}, {"title": "Segment CNN - Temporal Action Localization", "url": "https://www.ee.columbia.edu/ln/dvmm/researchProject/s/cdc/scnn.html"}, {"title": "Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/7780488/"}, {"title": "Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs", "url": "https://arxiv.org/pdf/1601.02129"}, {"title": "A two-stage temporal proposal network for precise action localization in untrimmed video | International Journal of Machine Learning and Cybernetics", "url": "https://link.springer.com/article/10.1007/s13042-021-01301-z"}, {"title": "GitHub - livst/scnn: Segment-CNN: A Framework for Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs", "url": "https://github.com/livst/scnn"}, {"title": "Action Temporal Localization in Untrimmed Videos via Multi-stage CNNs", "url": "https://www.researchgate.net/publication/290263840_Action_Temporal_Localization_in_Untrimmed_Videos_via_Multi-stage_CNNs"}, {"title": "Learning to Localize Actions from Moments", "url": "https://arxiv.org/pdf/2008.13705"}, {"title": "Temporal Context Network for Activity Localization in Videos", "url": "https://arxiv.org/pdf/1708.02349"}, {"title": "Graph Convolutional Module for Temporal Action Localization in Videos", "url": "https://arxiv.org/pdf/2112.00302"}]

I found the paper you're looking for! Here are the key details:

Title: Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs

Authors: Zheng Shou, Dongang Wang, and Shih-Fu Chang

Conference: CVPR 2016

Publication Details: Published at the IEEE conference on computer vision and pattern recognition (CVPR) in Las Vegas, NV, USA, 2016, pages 1049-1058

Overview: The paper proposes three segment-based 3D ConvNets: (1) a proposal network identifies candidate segments in a long video that may contain actions; (2) a classification network learns one-vs-all action classification model; and (3) a localization network fine-tunes the learned classification network to localize each action instance.

Key Innovation: The paper proposes a novel loss function for the localization network to explicitly consider temporal overlap and achieve high temporal localization accuracy.

Results: The approach achieves significantly superior performance: mAP increases from 1.7% to 7.4% on MEXaction2 and increases from 15.0% to 19.0% on THUMOS 2014.

The paper is also available on ArXiv and has an official implementation repository on GitHub.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-13T20:19:38.859Z)

Web search results for query: "Cascaded Boundary Regression for Temporal Action Detection Gao Yang Nevatia BMVC 2017"

Links: [{"title": "Cascaded Boundary Regression for Temporal Action Detection", "url": "https://bmva-archive.org.uk/bmvc/2017/papers/paper052/index.html"}, {"title": "Boundary-Aware Cascade Networks for Temporal Action Segmentation | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-030-58595-2_3"}, {"title": "[PDF] Cascaded Boundary Regression for Temporal Action Detection | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Cascaded-Boundary-Regression-for-Temporal-Action-Gao-Yang/34a468c0f6e299db7ca1228726f161856a3082ec"}, {"title": "Cascaded Boundary Regression for Temporal Action Detection - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2017arXiv170501180G/abstract"}, {"title": "Fast"}]

Learning of Temporal Action Proposal via Dense Boundary Generator", "url": "https://arxiv.org/pdf/1911.04127"}, {"title": "Rethinking Online Action Detection in Untrimmed Videos: A Novel Online Evaluation Protocol", "url": "https://arxiv.org/pdf/2003.12041"}, {"title": "[1705.01180] Cascaded Boundary Regression for Temporal Action Detection", "url": "https://arxiv.org/abs/1705.01180"}, {"title": "CASCADED BOUNDARY REGRESSION FOR TEMPORAL ACTION DETECTION 1", "url": "https://arxiv.org/pdf/1705.01180"}, {"title": "dblp: Cascaded Boundary Regression for Temporal Action Detection.", "url": "https://dblp.org/rec/journals/corr/GaoYN17.html"}, {"title": "ActionSpotter: Deep Reinforcement Learning Framework for Temporal Action Spotting in Videos", "url": "https://arxiv.org/pdf/2004.06971"}]

I found the paper you were looking for! Here are the details:

****Cascaded Boundary Regression for Temporal Action Detection**** by Jiyang Gao, Zhenheng Yang and Ram Nevatia, published at the British Machine Vision Conference (BMVC) in September 2017.

****Summary of the work:****

The paper presents a two-stage temporal action detection pipeline with a Cascaded Boundary Regression (CBR) model, where class-agnostic proposals and specific actions are detected respectively in the first and second stage. The key aspect of the refinement process is that the temporal boundaries are adjusted in a cascaded way by feeding the refined windows back to the system for further boundary refinement.

****Available resources:****

- The official BMVC archive version is available at: <https://bmva-archive.org.uk/bmvc/2017/papers/paper052/index.html>
- The ArXiv preprint is available at: <https://arxiv.org/abs/1705.01180>

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-13T20:19:42.826Z)

Both papers 1 and 2 are verified real. Note: paper 1's claimed arXiv ID is 1601.02129 ? confirmed. The S-CNN result number cited in the survey ("mAP at temporal IoU") is fine. Let me verify papers 3, 4, and 5.

7. user (2026-06-13T20:19:53.040Z)

Web search results for query: "Decoupled Classification Refinement Hard False Positive Suppression Object Detection Cheng Wei Feris Shi"

Links: [{"title": "[PDF] Decoupled Classification Refinement: Hard False Positive Suppression for Object Detection | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Decoupled-Classification-Refinement:-Hard-False-for-Cheng-Wei/af95ba2c08cf3317291dad554488dd951cd6ff80"}, {"title": "[1810.04002] Decoupled Classification Refinement: Hard False Positive Suppression for Object Detection", "url": "https://arxiv.org/abs/1810.04002"}, {"title": "Hard False Positive Suppression for Object Detection", "url": "https://arxiv.org/pdf/1810.04002"}, {"title": "GitHub - yinli0208/Decoupled-Classification-Refinement: The is an official implementation of our ECCV2018 paper \"Revisiting RCNN: On Awakening the Classification Power of Faster RCNN (https://arxiv.org/abs/1803.06799)\"", "url": "https://github.com/yinli0208/Decoupled-Classification-Refinement"}, {"title": "Decoupled Classification Refinement: Hard False Positive Suppression for Object Detection | Request PDF", "url": "https://www.researchgate.net/publication/328189677_Decoupled_Classification_Refinement_Hard_False_Positive_Suppression_for_Object_Detection"}, {"title": "GitHub - SHI-Labs/Decoupled-Classification-Refinement: Revisiting RCNN: On Awakening the Classification Power of Faster RCNN (ECCV 2018) · GitHub", "url": "https://github.com/SHI-Labs/Decoupled-Classification-Refinement"}, {"title": "Revisiting RCNN: On Awakening the Classification Power of Faster RCNN | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-030-01267-0_28"}, {"title": "A Global to Local Double Embedding Method for Multi-person Pose"}]

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Estimation","url":"https://arxiv.org/pdf/2102.07318"}, {"title": "Utilizing the Instability in Weakly Supervised Object Detection", "url": "https://arxiv.org/pdf/1906.06023"}]
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Based on the search results, here's what I found about this paper:

****Decoupled Classification Refinement: Hard False Positive Suppression for Object Detection**** is a paper by Bowen Cheng, Yunchao Wei, Rogerio Feris, Jinjun Xiong, Wen-mei Hwu, Thomas Huang, and Humphrey Shi.

****Main Contribution:****

The paper analyzes failure cases of state-of-the-art detectors and observes that most hard false positives result from classification instead of localization and they have a large negative impact on the performance of object detectors.

****Key Findings:****

The authors conjecture three factors: (1) Shared feature representation is not optimal due to the mismatched goals of feature learning for classification and localization; (2) multi-task learning helps, yet optimization of the multi-task loss may result in sub-optimal for individual tasks; (3) large receptive field for different scales leads to redundant context information for small objects.

****Proposed Solution:****

The paper demonstrates the potential of detector classification power by a simple, effective, and widely-applicable Decoupled Classification Refinement (DCR) network, which places a separate classification network in parallel with the localization network (base detector). DCR samples hard false positives from the base classifier in Faster RCNN and trains a RCNN-styled strong classifier.

The paper was published as an arXiv preprint (arXiv:1810.04002) in 2018.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-13T20:19:53.717Z)

Web search results for query: "Rapid Object Detection using a Boosted Cascade of Simple Features Viola Jones CVPR 2001"

Links: [{"title": "P. Viola and M. Jones, ?Rapid Object Detection Using a Boosted Cascade of Simple Features,? Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR), Vol. 1, 2001, pp. 511-518. - References - Scientific Research Publishing", "url": "https://www.scirp.org/reference/referencespapers?referenceid=953018"}, {"title": "Viola, P. and Jones, M. (2001) Rapid Object Detection Using a Boosted Cascade of Simple Features. Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition. CVPR 2001, Kauai, 8-14 December 2001, I-511-I-518. - References - Scientific Research Publishing", "url": "https://www.scirp.org/reference/referencespapers?referenceid=2961550"}, {"title": "[PDF] Rapid object detection using a boosted cascade of simple features | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Rapid-object-detection-using-a-boosted-cascade-of-Viola-Jones/dc6ea0e30e46163b706f2f8bdc9c67ca87f83d63"}, {"title": "(PDF) Rapid object detection using a boosted cascade of simple features", "url": "https://www.academia.edu/96737309/Rapid_object_detection_using_a_boosted_cascade_of_simple_features"}, {"title": "Rapid Object Detection Using a Boosted Cascade of Simple Features citation", "url": "https://duruofoei.com/cites/Viola2001Rapid.html"}, {"title": "Vision based real time traffic monitoring", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/8379926"}, {"title": "(PDF) Rapid Object Detection using a Boosted Cascade of Simple Features", "url": "https://www.researchgate.net/publication/3940582_Rapid_Object_Detection_using_a_Boosted_Cascade_of_Simple_Features"}, {"title": "Method and system for detecting object", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/8520893"}, {"title": "Supervised Transformer Network for Efficient Face Detection", "url": "https://arxiv.org/pdf/1607.05477"}, {"title": "Object detecting apparatus and learning apparatus for the same", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/8233720"}]

Based on the search results, here's what I found about the paper "Rapid Object Detection Using a

Boosted Cascade of Simple Features":

Paper Details

The paper was authored by Viola and Jones and presented at the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2001), held in Kauai from December 8-14, 2001, with pages I-511-I-518.

Key Contributions

This work describes a machine learning approach for visual object detection which is capable of processing images extremely rapidly and achieving high detection rates. The paper made three significant contributions:

1. The introduction of a new image representation called the "integral image" which allows the features used by the detector to be computed very quickly.
2. Viola and Jones introduced the first real-time face detector by combining Haar-like features with AdaBoost-trained cascade classifiers.
3. Their integral image representation allowed rapid feature computation, and the cascade structure rejected non-face regions early, concentrating computation on promising candidates.

Citation Information

The paper's DOI is 10.1109/CVPR.2001.990517, making it easy to locate and reference in academic databases.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-13T20:19:59.521Z)

Papers 3, 4 verified. Note one nuance on paper 3 (DCR): the survey lists the author "Honghui Shi" but the actual paper lists "Humphrey Shi" (Honghui Shi is the same person ? Humphrey is the English name for Honghui Shi). So this is correct, not a fabrication. The DOI for Viola-Jones is confirmed (10.1109/CVPR.2001.990517). Let me verify papers 5, 6, 7, 8.

10. user (2026-06-13T20:21:15.371Z)

Web search results for query: "Enhancing Badminton Game Analysis Shot Refinement Fusion Shuttlecock Tracking Hit Detection Monocular Sensors 2024 Hsu Yu Cheng"

Links: [{"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Fine reconstruction of badminton swing dynamic trajectory assisted by event camera | Scientific Reports", "url": "https://www.nature.com/articles/s41598-026-46443-8"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title": "(PDF) Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.researchgate.net/publication/382039345_Enhancing_Badminton_Game_Analysis_An_Approach_to_Shot_Refinement_via_a_Fusion_of_Shuttlecock_Tracking_and_Hit_Detection_from_Monocular_Camera"}, {"title": "A Model for Analyzing Shuttlecock Landing Position Using Real-Time Object Detection Techniques | Journal of Applied Science and Emerging Technology", "url": "https://ph01.tci-thaijo.org/index.php/JASCI/article/view/265289"}, {"title": "Future Prediction of Shuttlecock Trajectory in Badminton Using Player's Information - PubMed", "url": "https://pubmed.ncbi.nlm.nih.gov/37233318/"}, {"title": "Use predictive analytics and big data technologies to enhance badminton game strategy development and performance indicators - Xin Feng, Xiang Liu,"}

2025", "url": "https://journals.sagepub.com/doi/10.1177/14727978251324125"}, {"title": "MATCHMOTION: AN AI-POWERED MOBILE APPLICATION FOR AUTOMATED BADMINTON VIDEO ANALYSIS USING YOLOV8 OBJECT DETECTION", "url": "https://www.researchgate.net/publication/404597971_MATCHMOTION_AN_AI-POWERED_MOBILE_APPLICATION_FOR_AUTOMATED_BADMINTON_VIDEO_ANALYSIS_USING_YOLOV8_OBJECT_DETECTION"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"}, {"title": "AI-Driven Badminton Analysis: Enhancing Shot Detection and Player Performance Strategies | Technology", "url": "https://www.devdiscourse.com/article/technology/3009448-ai-driven-badminton-analysis-enhancing-shot-detection-and-player-performance-strategies"}]]

API Error: The socket connection was closed unexpectedly. For more information, pass `verbose: true` in the second argument to fetch()

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-13T20:21:15.989Z)

Web search results for query: "TennisExpert Towards Expert-Level Analytical Sports Video Understanding arXiv 2026"

Links: [{"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/html/2603.13397v2"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "BoxComm: Benchmarking Category-Aware Commentary Generation and Narration Rhythm in Boxing", "url": "https://arxiv.org/pdf/2604.04419"}, {"title": "(PDF) TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://www.researchgate.net/publication/402480116_TennisExpert_Towards_Expert-Level_Analytical_Sports_Video_Understanding"}, {"title": "TenniSet: A Dataset for Dense Fine-Grained Event Recognition, Localisation and Description", "url": "https://www.researchgate.net/publication/322003612_TenniSet_A_Dataset_for_Dense_Fine-Grained_Event_Recognition_Localisation_and_Description"}, {"title": "GitHub - LZYAndy/TennisExpert · GitHub", "url": "https://github.com/LZYAndy/TennisExpert"}, {"title": "[2507.02906] Enhancing Sports Strategy with Video Analytics and Data Mining: Automated Video-Based Analytics Framework for Tennis Doubles", "url": "https://arxiv.org/abs/2507.02906"}]]

API Error: The socket connection was closed unexpectedly. For more information, pass `verbose: true` in the second argument to fetch()

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-13T20:21:21.570Z)

Both papers 5 and 6 verified real. Important nuance on paper 5: the survey gives the title as "Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from a Monocular Camera" but the actual title is "Enhancing Badminton Game Analysis: **An Approach to** Shot Refinement via **a** Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera." Minor paraphrase, same paper, DOI 10.3390/s24134372 and PMC11244353 both confirmed.

Paper 6 (TennisExpert): arXiv:2603.13397 confirmed real, with a GitHub repo. The "2026" year matches the arXiv ID prefix (2603 = March 2026). Now let me verify papers 7 and 8, plus the ASRF paper referenced in techniques.

13. user (2026-06-13T20:21:27.677Z)

Web search results for query: "'A New Perspective for Shuttlecock Hitting Event Detection' Yu-Hsi Chen arXiv 2306.10293"

Links: [{"title": "A New Perspective for Shuttlecock Hitting Event Detection", "url": "https://arxiv.org/pdf/2306.10293"}, {"title": "[2306.10293] A New Perspective for Shuttlecock Hitting Event Detection", "url": "https://arxiv.org/abs/2306.10293"}, {"title": "A New Perspective for Shuttlecock Hitting Event Detection", "url": "https://arxiv.org/html/2306.10293"}, {"title": "A-New-Perspective-for-Shuttlecock-Hitting-Event-Detection/README.md at main · wish44165/A-New-Perspective-for-Shuttlecock-Hitting-Event-Detection", "url": "https://github.com/wish44165/A-New-Perspective-for-Shuttlecock-Hitting-Event-Detection/blob/main/README.md"}, {"title": "(PDF) A New Perspective for Shuttlecock Hitting Event Detection", "url": "https://www.researchgate.net/publication/371728085_A_New_Perspective_for_Shuttlecock_Hitting_Event_Detection"}, {"title": "Paper page - A New Perspective for Shuttlecock Hitting Event Detection", "url": "https://huggingface.co/papers/2306.10293"}, {"title": "GitHub - wish44165/A-New-Perspective-for-Shuttlecock-Hitting-Event-Detection: ? IJCAI 2023 (Macao, S.A.R) - IT4PSS Workshop ?", "url": "https://github.com/wish44165/A-New-Perspective-for-Shuttlecock-Hitting-Event-Detection"}, {"title": "Papers with Code - A New Perspective for Shuttlecock Hitting Event Detection", "url": "https://paperswithcode.com/paper/a-new-perspective-for-shuttlecock-hitting"}]

I found the paper! Here's information about it:

Title: A New Perspective for Shuttlecock Hitting Event Detection

Author: Yu-Hsi Chen from the Institute of Communications Engineering, National Tsing Hua University, Taiwan

Key Approach: The paper captures the hitting action of players by reasoning over a sequence of images, utilizing a deep learning model called SwingNet. This model is designed to capture the relevant characteristics and patterns associated with the act of hitting in badminton.

Methodology: The proposed method applies a specific video processing technique to extract prior features from the video, which significantly reduces the learning difficulty for the model.

Availability: The source code is available at <https://github.com/TW-yuhsi/A-New-Perspective-for-Shuttlecock-Hitting-Event-Detection>.

The paper was published on arXiv on June 17, 2023, and presents a novel computer vision approach for automatically detecting shuttlecock hitting events in badminton match videos.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-13T20:21:29.645Z)

Web search results for query: "Alleviating Over-segmentation Errors by Detecting Action Boundaries ASRF Ishikawa 2020"

Links: [{"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/iel7/9423008/9423028/09423184.pdf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | DeepAI", "url": "https://deepai.org/publication/alleviating-over-segmentation-errors-by-detecting-action-boundaries"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://openaccess.thecvf.com/content/WACV2021/papers/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.pdf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request PDF", "url": "https://www.researchgate.net/publication/342944719_Alleviating_Over-segmentation_Errors_by_Detecting_Action_Boundaries"}, {"title": "[PDF] Alleviating Over-segmentation Errors by Detecting Action Boundaries | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Alleviating-Over-segmentation-Errors-by-Detecting-Ishikawa-Kasai/4fed0235614df46fa0c53e05b71e9939b53a1bdf"}, {"title": "Alleviating Over-Segmentation Errors by Detecting Action"}]

```
Boundaries", "url": "https://papertalk.org/papertalks/27315"}, {"title": "GitHub - yiskw713/asrf: Code for 'Alleviating Over-segmentation Errors by Detecting Action Boundaries' accepted in WACV2021", "url": "https://github.com/yiskw713/asrf"}, {"title": "Permutation-Aware Action Segmentation via Unsupervised Frame-to-Segment Alignment", "url": "https://arxiv.org/pdf/2305.19478"}]
```

Based on the search results, I found information about this paper:

The paper proposes an Action Segment Refinement Framework (ASRF) that consists of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB).

The long-term feature extractor provides shared features for the two branches with a wide temporal receptive field, with the ASB classifying video frames with action classes, while the BRB regresses the action boundary probabilities. The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement.

The ASB and the BRB are complementary to each other, which enables the reduction of over-segmentation errors.

The paper was accepted in WACV 2021, and the authors are Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, and Hirokatsu Kataoka from the National Institute of Advanced Industrial Science and Technology (AIST).

The official implementation code is available on GitHub at: <https://github.com/yiskw713/asrf>

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-06-13T20:21:36.399Z)

All papers 1-7 are verified real. The ASRF reference cited in techniques is also real (WACV 2021, arXiv:2007.06866 ? minor note: survey says "2020" but it's WACV 2021 / arXiv mid-2020, so the year is essentially accurate as a preprint).

Now for the adversarial part: the verdict is "partial-overlap." This isn't a "novel-combination" or "genuine-gap" verdict, so I'm not strictly required to refute it ? but I should sanity-check whether the verdict should be MORE toward "reinventing" (the more skeptical direction). The survey itself argues the individual moves are all established and only the packaging is novel. Let me probe two things: (1) whether the "off-policy evaluation" framing reference (arXiv:2103.16596) is real, and (2) whether there's even closer prior art that does the specific "persist perception once, re-score offline" + feature-caching combination in video ? which would push toward "reinventing."

16. user (2026-06-13T20:21:44.465Z)

Web search results for query: "Benchmarks for Deep Off-Policy Evaluation" arXiv 2103.16596"

Links: [{"title": "Benchmarks for Deep Off-Policy Evaluation - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2021arXiv210316596F/abstract"}, {"title": "[2103.16596v1] Benchmarks for Deep Off-Policy Evaluation", "url": "https://arxiv.org/abs/2103.16596v1"}, {"title": "benchmarks for deep off-policy evaluation", "url": "https://arxiv.org/pdf/2103.16596"}, {"title": "[PDF] Benchmarks for Deep Off-Policy Evaluation | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Benchmarks-for-Deep-Off-Policy-Evaluation-Fu-Norouzi/d86dfdbb8eab91cf23b81c541b4f741f88b7d756"}, {"title": "Benchmarks for Deep Off-Policy Evaluation | DeepAI", "url": "https://deepai.org/publication/benchmarks-for-deep-off-policy-evaluation"}, {"title": "dblp: Benchmarks for Deep Off-Policy Evaluation.", "url": "https://dblp.org/rec/conf/iclr/Fu0NTw0YZCKPLP21.html"}, {"title": "SOPE: Spectrum of Off-Policy Estimators", "url": "https://arxiv.org/pdf/2111.03936"}, {"title": "Benchmarks for Deep Off-Policy Evaluation | OpenReview", "url": "https://openreview.net/forum?id=kWSeGEeHvF8"}, {"title": "Showing Your Offline Reinforcement Learning Work: Online Evaluation Budget Matters", "url": "https://arxiv.org/pdf/2110.04156"}, {"title": "(PDF) Benchmarks for Deep Off-Policy Evaluation", "url": "https://www.academia.edu/73389071/Benchmarks_for_Deep_Off_Policy_Evalu"}]

ation"]}]

I found information about the paper "Benchmarks for Deep Off-Policy Evaluation" (arXiv:2103.16596). Here are the key details:

****Paper Overview:****

The paper was submitted on March 30, 2021, and is authored by Justin Fu, Mohammad Norouzi, Ofir Nachum, George Tucker, Ziyu Wang, Alexander Novikov, Mengjiao Yang, Michael R. Zhang, Yutian Chen, Aviral Kumar, Cosmin Paduraru, Sergey Levine, and Tom Le Paine. It was published as an ICLR 2021 paper.

****Purpose:****

Off-policy evaluation (OPE) holds the promise of being able to leverage large, offline datasets for both evaluating and selecting complex policies for decision making. This is particularly important in real-world domains such as healthcare, recommender systems, or robotics, where online data collection is expensive and potentially dangerous, and could yield significant benefits in safety, time, and cost.

****Main Contribution:****

The paper's primary contribution is the Deep Off-Policy Evaluation (DOPE) benchmark. The goal is to provide a set of tasks with a range of difficulty that exercise various design properties and provide policies with different behavioral patterns to establish a standardized framework for comparing OPE algorithms, with particular emphasis on large datasets, long-horizon tasks, and task complexity.

****Resources:****

Policies and evaluation code are available at https://github.com/google-research/deep_ope

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

17. user (2026-06-13T20:21:56.867Z)

Web search results for query: "precompute cache video features frozen backbone offline re-scoring temporal action localization feature store reuse"

Links: [{"title": "RCL: Recurrent Continuous Localization for Temporal Action Detection", "url": "https://arxiv.org/pdf/2203.07112"}, {"title": "Revisiting Feature Prediction for Learning Visual Representations from Video", "url": "https://arxiv.org/pdf/2404.08471"}, {"title": "TallFormer: Temporal Action Localization with a Long-memory Transformer", "url": "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136940495.pdf"}, {"title": "Exploring Temporal Granularity in Self-Supervised Video Representation Learning", "url": "https://arxiv.org/pdf/2112.04480"}, {"title": "PcmNet: Position-Sensitive Context Modeling Network for Temporal Action Localization", "url": "https://arxiv.org/pdf/2103.05270"}, {"title": "Exploring Stronger Feature for Temporal Action Localization", "url": "https://arxiv.org/pdf/2106.13014"}, {"title": "TimeLoc: A Unified End-to-End Framework for Precise Timestamp Localization in Long Videos", "url": "https://arxiv.org/html/2503.06526v1"}, {"title": "OpenTAD: A Unified Framework and Comprehensive Study of ...", "url": "https://openaccess.thecvf.com/content/CVPR2025W/PVUW/papers/Liu_OpenTAD_A_Unified_Framework_and_Comprehensive_Study_of_Temporal_Action_CVPRW_2025_paper.pdf"}, {"title": "LoSA: Long-Short-range Adapter for Scaling End-to-End Temporal Action Localization", "url": "https://arxiv.org/html/2404.01282"}, {"title": "End-to-End Temporal Action Detection with 1B Parameters Across 1000 Frames", "url": "https://arxiv.org/html/2311.17241v2"}]

I found several relevant research papers on temporal action localization. Here are the key findings:

Video Features and Frozen Backbones

Video features are commonly offline extracted from untrimmed video frames using 3D convolutional networks for temporal action detection approaches. Many approaches use off-the-shelf video recognition models with frozen parameters of the video feature extractor for training efficiency.

Training Strategies with Frozen Backbones

The search results reveal several approaches to temporal action localization:

- Head-only transfer learning involves processing untrimmed video frames as independent clips by a frozen backbone, concatenating features after the last layer, and feeding them to a learnable TAL head.
- When feature extraction backbones are frozen, frame representation yields superior results to snippet representation for both VideoMAE and SlowFast backbones.

Re-scoring and Feature Reuse

A typical pipeline involves Stage 0 where videos are encoded into features using a pretrained video backbone (either fine-tuned or frozen), followed by Stage 1 which consists of a neck for temporal aggregation of snippet-level features and a dense head that generates snippet-level predictions.

Recent Advances

Recent research demonstrates that in temporal action localization, efficiently scaling input frames and resolution under end-to-end learning yields better performance than relying on frozen features, though frozen backbone approaches remain computationally practical.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. assistant (2026-06-13T20:22:05.681Z)

This is significant for the adversarial check. The off-policy benchmark reference (arXiv:2103.16596) is real. More importantly, the second search confirms the survey's own honesty: "frozen-backbone precomputed (cached) features" + "Stage 0 encode ? Stage 1 head re-score" is the **standard, dominant practice** in the entire temporal action localization (TAL) field, not a novel substrate. The "compute features once with a frozen backbone, then train/re-score a cheap head many times" is exactly how the whole TAD literature operates (this is why benchmarks like THUMOS/ActivityNet ship precomputed I3D/TSN/VideoMAE features).

This pushes the verdict question: the survey claims the novelty is in the **combination** ? (a) persist-once/re-score-many substrate, (b) "signal not verdict" feature-column discipline, (c) report-first reconciler linter. But (a) is literally the field's default. Let me probe whether the remaining distinctive packaging (the LR-scorer-over-cached-cues for racket-sport rally segmentation + report-first reconcilers) has even closer prior art that would make this more "reinventing" than "partial-overlap." Let me also confirm the closest racket-sport pipeline shape isn't even closer than claimed.

19. user (2026-06-13T20:22:14.502Z)

Web search results for query: "sports broadcast rally point segmentation lightweight classifier cached per-frame features leave-one-video-out temporal IoU"

Links: [{"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos", "url": "https://arxiv.org/pdf/2002.08097"}, {"title": "Video Segmentation in Computer Vision: A Short Guide [+Examples]", "url": "https://www.lightly.ai/blog/video-segmentation"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "Precise Event Spotting in Sports Videos: Solving Long-Range Dependency and Class Imbalance", "url": "https://arxiv.org/html/2503.00147v1"}, {"title": "GSVNet: Guided Spatially-Varying Convolution for Fast Semantic Segmentation on Video", "url": "https://arxiv.org/pdf/2103.08834"}, {"title": "SoccerNet-v2: A Dataset and Benchmarks for Holistic Understanding of Broadcast Soccer"}]

Videos", "url": "https://arxiv.org/pdf/2011.13367"}, {"title": "Temporal Tampering Detection in Automotive Dashcam Videos via Multi-Feature Forensic Analysis and a 1D Convolutional Neural Network", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12846185/"}, {"title": "SmartTennisTV: Automatic indexing of tennis videos", "url": "https://arxiv.org/pdf/1801.01430"}]

The search returned results related to sports broadcast video analysis, particularly focusing on rally segmentation in tennis and badminton videos. Here are the key findings:

****Rally Segmentation in Sports Broadcasts:****

A lightweight, purely audio-based detection model can be employed to identify the distinct acoustic signatures of racquet-ball impacts from the broadcast audio track for efficient rally segmentation across long videos.

****Classification Approaches:****

Rather than using HOG + SVM, a convolutional neural network based classifier such as ResNet-18 (pre-trained on ImageNet) can be used to classify rally and non-rally frames. This CNN-based method is both more accurate and exhibits lower latency compared to HOG + SVM.

****Frame-Level Features and Evaluation:****

Frame-level features can be compared to player-level features for evaluation, and player-level features generalize better compared to global frame-level features.

****Temporal Metrics:****

In testing, a sliding window approach with stride of 1 frame can be used to infer scores in time, reduced to predictions using NMS, which relates to the temporal evaluation methodologies used in sports video analysis.

The search results show active research in sports broadcast video understanding, particularly for automatic rally detection and segmentation tasks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

20. user (2026-06-13T20:22:18.378Z)

Web search results for query: "badminton rally segmentation play break detection logistic regression features shuttle trajectory person audio cues"

Links: [{"title": "Future Prediction of Shuttlecock Trajectory in Badminton Using Player's Information", "url": "https://www.mdpi.com/2313-433X/9/5/99"}, {"title": "Future Prediction of Shuttlecock Trajectory in Badminton Using Player's Information", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10219238/"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/pdf/2306.04948"}, {"title": "Benchmarking Badminton Action Recognition with a New Fine-Grained Dataset", "url": "https://arxiv.org/html/2403.12385v2"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title": "Bridging the Gap: Doubles Badminton Analysis with Singles-Trained Models", "url": "https://arxiv.org/html/2508.13507v1"}]

Based on the web search results, I found several relevant research articles on badminton rally segmentation and video analysis:

Rally Segmentation and Detection

Rally segmentation involves annotators segmenting the start and end of each rally and the scoring result by recording the time of the corresponding event, which is supported by the shuttle trajectory. Video preprocessing splits the match video into multiple rally videos to avoid heavily wasting computing resources and to eliminate execution time due to the break time.

Key Features and Methods

The research uses multiple feature types for badminton analysis:

- **Shuttle Trajectory**: Detecting the shuttlecock's trajectory is used to aid in microscopic labeling where each shot is analyzed in detail. TrackNet, a deep neural network designed for tracking small objects, is used to extract the flight trajectory of the shuttlecock.

- **Player Information**: Methods take into account both the shuttlecock position and the positions and postures of the players. The YOLOv7 model is used to identify whether the player is swinging.

- **Shot Detection**: Systems are capable of detecting the trajectory of the shuttlecock, thereby enabling the identification of the players' swing moments, segmentation of each round, summarization of points won and lost, and categorization of the shots.

The search results focus primarily on computer vision-based approaches using shuttle tracking and player pose detection, though the specific mention of logistic regression features and audio cues was not found in these particular results.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. assistant (2026-06-13T20:22:26.224Z)

These searches surface two additional relevant prior-art items the survey didn't cite, which are worth flagging as course corrections but don't fundamentally change the verdict:

- **"Towards Real-Time Analysis of Broadcast Badminton Videos"** (arXiv:2308.12199) ? broadcast badminton rally analysis pipeline.

- **"SmartTennisTV: Automatic indexing of tennis videos"** (arXiv:1801.01430) ? rally-level indexing of tennis broadcasts (rally/non-rally frame classification ? segmentation), which is conceptually close to the rally-indexing decision layer.

Neither does the exact "persist cheap LR scorer over cached multi-cue features + report-first reconcilers, LOVO-graded" combination, so they reinforce "partial-overlap" rather than push to "reinventing." The survey's self-assessment is honest and well-calibrated.

Let me do one final probe to make sure I'm not missing the single closest match: a paper that explicitly does "cue feature columns ? learned scorer (not if/else) for rally/play-break segmentation," which is the survey's claimed distinctive (b).

22. user (2026-06-13T20:22:32.925Z)

Web search results for query: "SmartTennisTV automatic indexing tennis videos rally segmentation Vinay Nambodiri"

Links: [{"title": "[1801.01430] SmartTennisTV: Automatic indexing of tennis videos", "url": "https://arxiv.org/abs/1801.01430"}, {"title": "SmartTennisTV: Automatic indexing of tennis videos | DeepAI", "url": "https://deepai.org/publication/smarttennistv-automatic-indexing-of-tennis-videos"}, {"title": "SmartTennisTV: Automatic indexing of tennis videos", "url": "https://cvit.iiit.ac.in/research/projects/cvit-projects/smart-tennistv"}, {"title": "SmartTennisTV: Automatic indexing of tennis videos Anurag Ghosh and C.V.Jawahar", "url": "https://cdn.iiit.ac.in/cdn/cvit.iiit.ac.in/images/ConferencePapers/2017/smarttennistv-automatic-indexing.pdf"}, {"title": "(PDF) SmartTennisTV: Automatic indexing of tennis videos", "url": "https://www.researchgate.net/publication/322305899_SmartTennisTV_Automatic_indexing_of_tennis_videos"}, {"title": "SmartTennisTV: Automatic Indexing of Tennis Videos | SpringerLink", "url": "https://link.springer.com/chapter/10.1007/978-981-13-0020-2_3"}, {"title": "SmartTennisTV: Automatic indexing of tennis"}]

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videos", "url": "https://arxiv.org/pdf/1801.01430"}, {"title": "SmartTennisTV: Automatic Indexing of Tennis Videos | Request PDF", "url": "https://www.researchgate.net/publication/324770372_SmartTennisTV_Automatic_Indexing_of_Tennis_Videos"}, {"title": "[PDF] SmartTennisTV: Automatic indexing of tennis videos | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/SmartTennisTV:-Automatic-indexing-of-tennis-videos-Ghosh-Jawahar/6e926ea795c320389392183a7f348ef36c344b18"}, {"title": "Frame level annotations for tennis videos", "url": "https://www.researchgate.net/publication/316448127_Frame_level_annotations_for_tennis_videos"}]
```

Based on the search results, I found information about SmartTennisTV, though I should note that the paper appears to be authored by Anurag Ghosh and C.V. Jawahar from IIIT Hyderabad, not Vinay Namboodiri. Here's what the research covers:

SmartTennisTV demonstrates a score-based indexing approach for tennis videos that indexes video segments with their scores to create a navigable and searchable match.

The approach temporally segments the rallies in the video and then recognizes the scores from each of the segments, before refining the scores using the knowledge of the tennis scoring system. The system builds an interface to effortlessly retrieve and view relevant video segments by automatically tagging the segmented rallies with human accessible tags such as 'fault' and 'deuce'.

The efficiency of the approach is demonstrated on broadcast tennis videos from two major tennis tournaments.

If you were looking for work specifically by Vinay Namboodiri related to tennis video analysis, you may want to verify the authorship, as the search results consistently show this work as being published by Ghosh and Jawahar at IIIT Hyderabad.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

23. user (2026-06-13T20:22:36.914Z)

Web search results for query: "play-break segmentation sports video machine learning classifier features tennis badminton volleyball "leave-one-video-out" evaluation"

Links: [{"title": "Automatic Detection of Play and Break Segments in Basketball Videos Based on the Analysis of the Slope of the Basketball Court Boundary | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-030-28374-2_55"}, {"title": "Method and system for high-level structure analysis and event detection in domain specific videos", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/6813313"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "TrackNet: A Deep Learning Network for Tracking High-speed and Tiny Objects in Sports Applications", "url": "https://arxiv.org/pdf/1907.03698"}, {"title": "Sports Video Classification Method Based on Improved Deep Learning", "url": "https://www.mdpi.com/2076-3417/14/2/948"}, {"title": "(PDF) Classification of sports videos with combination of deep learning models and transfer learning", "url": "https://www.researchgate.net/publication/332582310_Classification_of_sports_videos_with_combination_of_deep_learning_models_and_transfer_learning"}, {"title": "Sports video retrieval and classification using focus u-net based squeeze excitation and residual mapping deep learning model - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/abs/pii/S095219762301360X"}, {"title": "Sports Video Classification Framework Using Enhanced Threshold Based Keyframe Selection Algorithm and Customized CNN on UCF101 and Sports1-M Dataset", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9754837/"}]

I found search results related to sports video analysis and machine learning classification, though the specific paper you're looking for with that exact combination of features doesn't appear in these results. However, I found some highly relevant research:

One study on badminton video analysis extracted HOG features from every 10th frame and learned a 2^2 kernel SVM classifier to label frames as "point frame" or "non-point frame", then smoothed the sequence using a Kalman filter. The F1 score for the classifier achieved 95.44%, with

precision and recall for the point class at 97.83% and 91.02% respectively.

Other research on play-break detection in sports events used a context descriptor for real-time visual play-break detection, and there has been work on frameworks for flexible summarization of racquet sports video using multiple modalities.

The search results contain various studies on sports video classification covering tennis, badminton, and volleyball, but the specific research paper using the exact evaluation methodology you mentioned ("leave-one-video-out") with those particular features doesn't appear in these results. You may want to search academic databases like IEEE Xplore, ACM Digital Library, or Google Scholar directly for this specific combination of methods.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

24. assistant (2026-06-13T20:22:49.477Z)

These searches surface the closest prior-art family for the *decision layer* specifically, which strengthens the "established, cite-don't-claim" case:

- **"Towards Structured Analysis of Broadcast Badminton Videos"** (arXiv:1712.08714) ? extracts HOG features per frame, learns a γ^2 -kernel **SVM** classifier to label point/non-point frames, then **smooths** the sequence (Kalman) ? this is precisely the "cheap learned classifier over per-frame features ? temporal smoothing for rally/non-rally segmentation" decision layer, in the exact sport. It is arguably the single closest analog to the survey's "decision layer" claimed distinctiveness (b), and the survey did NOT cite it.
- **SmartTennisTV** (arXiv:1801.01430) ? temporally segments rallies then re-scores/refines using domain knowledge of scoring rules ? a "segment then re-score/refine with a rule pass" pipeline, close to the reconciler idea (c).

These don't do the *full* combination (cached multi-cue substrate + "signal-not-verdict" LR + report-first A/B-gated reconciler linter + LOVO/tIoU harness), so the survey's "partial-overlap" verdict still holds ? but they materially narrow the novelty gap and are the most important course corrections to add. I have enough to finalize.

Summary of verification: all 8 cited papers (plus the 2 extra references in the techniques section ? ASRF and the DOPE off-policy benchmark) are REAL. No fabrications. A few minor title paraphrases (paper 5's title) and one author-name nuance (paper 3: "Honghui Shi" = "Humphrey Shi", same person) ? none rise to fabrication. The verdict is well-calibrated; if anything it should stay partial-overlap, with a nudge in the skeptical direction because frozen-backbone feature caching is the field's default and there is uncited closer prior art for the decision layer.

25. user (2026-06-13T20:23:43.000Z)

Structured output provided successfully

26. assistant (2026-06-13T20:23:47.352Z)

Verification complete. All 8 cited papers plus the 2 supplementary references (ASRF, DOPE) are REAL ? zero fabrications. Verdict upheld as partial-overlap with a documented skeptical nudge: the "persist-once/re-score-many frozen-feature substrate" is the TAD field's default (not novel), and there is uncited same-sport prior art (arXiv:1712.08714) for the learned decision layer. The structured object has been returned.